Manufacture

"The SMT MARKET Is BOOMING But Operators Do Not Have Enough Knowledge About INSPECTION MACHINES"

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The continuing evolution of miniature packaging has led to ever-increasing PCB density and complexity. As manufacturing processes become progressively more complicated, there is a high probability of defects on finished PCB assemblies. Peter Shin, managing director, Koh Young SE Asia, in conversation with Baishakhi Dutta, shares how his organisation is making 3D inspection systems a viable technology for electronics manufacturing

Q. How are you handling the transition from 2D to 3D inspection technology?

A. 2D is based on grey-scale value comparison, whereas 3D is completely different. In 2009, we developed 3D SPI (solder paste inspection), and were the first ones in this market. Now, our SPI market share is 53 per cent worldwide. When one of our big EMS customers asked us to jump into the automated optical inspection (AOI) market, since he considered us to be the best in 3D SPI, we said no. We found that 2D inspection depends entirely on human beings or conditional changes.

Q. What is the greatest challenge that you face in the SMT equipment ecosystem?

A. The surface-mount technology (SMT) market is booming but operators do not have enough knowledge about inspection machines. The machines need to deliver a clear idea to them so that users can minimise their decision making. This also ensures better quality.

Q. What is the USP of your product lines?

A. We try to deliver real measurement values. 3D not only means x and y but also z dimension. Good machining can be performed in any circumstance or place. Some machines perform well in the US because they have ownership and in Europe because they study a lot about their processes. But, in China and South-East Asia, operators do not have ownership and never study about their job and tools. Thus, the global customer complains about the machines not working in places like India and China. Machines can tell only good work. With measurement data, it is easy. The machines can then be controlled globally.

Q. How are you upgrading the technology of your products to fit modern industry needs?

A. 3D measurement requires sensor technology. At Koh Young, we use fine measurement technology apart from using a tracking and semiconductor system. Industry 4.0, which is about automation of the factory production line, talks about the need of the factory to monitor incoming data, product schedule, any imbalance in incoming material and output. Everything needs to be synchronised so that the customer can quantify everything.

Q. Do you use any support tools to help your customers?

A. We have a solution called KSmart, which will soon have a second version. In the first version, we tried to collect all data and then display it. In the second version, data is not only shown but is also guided. When you see a defect or the result, you click and read the root codes. After you see the overall status, we specify where exactly you need to look. So, this is the first level. In the second level, we deliver more ideas and try to control other production machines with our data.

Our KPOL (Koh Young process optimisation tool) takes care of communication with the printer. It is based on artificial intelligence (AI) and is almost better than a human being because we task it with over 20 years of experience.

Q. Has the Indian government's policy against importing second-hand equipment affected your business?

A. We do not have any second-hand machines. The market price is now going down and the customer does not bother about whether it is first-hand or second-hand. We, as manufacturers, can only drop our selling price.

Second-hand machines are mostly 2D, and not 3D. I cannot say much about 3D machines, but the Chinese try to sell it at a lower price. So now we are fighting on the pricing part. In India, volume is not as big as in China, but price is almost the same. And operation force is even higher than that in China. So, I would say no, this policy is not affecting our business directly!

Q. What is the roadmap for the future?

A. That is a difficult question! Our hardware condition is quite stable. And now we are trying to focus on AI such as auto-programming, auto-finetuning and further enhancing KSmart. Apart from this, one of our major focus areas is M2M development as part of the smart factory solution. EFY